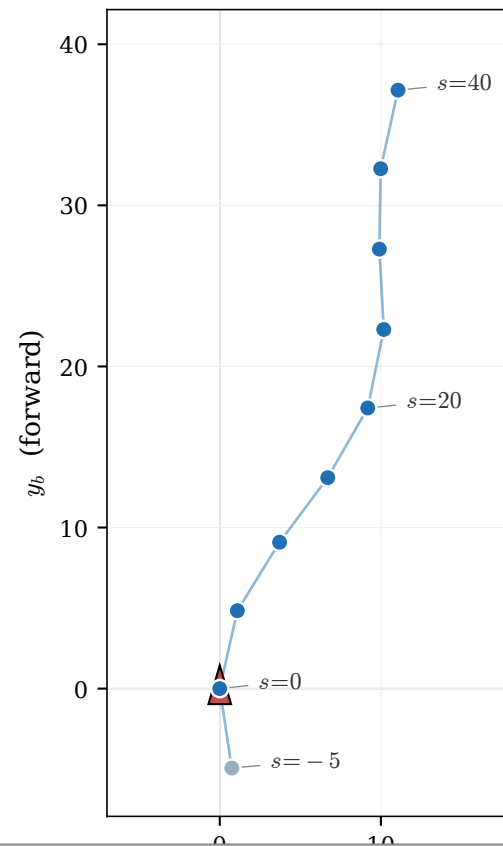


Lane waypoint extraction — built once per episode by `scripts/lane\_preprocess.py`

(b) Body view — the 20-D lane vector

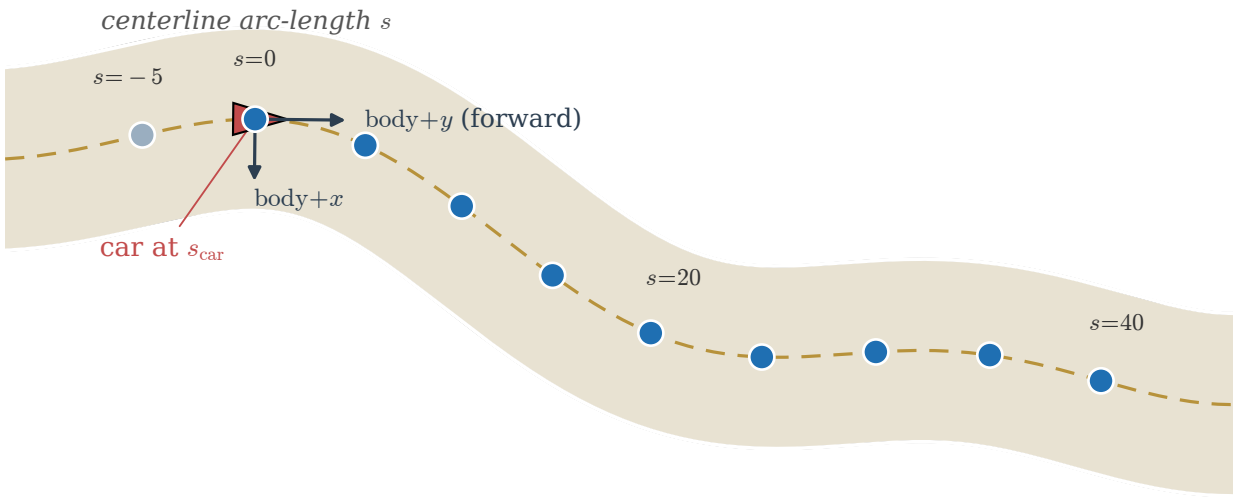
$$z^{\text{lane}} = (x_b^0, y_b^0, x_b^1, y_b^1, \dots, x_b^9, y_b^9)$$



Step 3 · World → body frame

$$w_i = R(-\text{yaw}) \cdot (w_i^{\text{world}} - \text{car_pos})$$

(a) World view — sample centerline at $\{s_{\text{car}} + s_i\}_{i=0}^9, s_i \in \{-5, 0, 5, \dots, 40\}$



Step 1 · Project car to centerline

$$s_{\text{car}} = \text{xy_to_sd}(\text{car_pos}) \quad (\text{Frenet } s\text{-coordinate})$$

Step 2 · Sample 10 fixed offsets

$$w_i^{\text{world}} = \text{sd_to_xy}(s_{\text{car}} + s_i, d=0), \quad s_i \in \{-5, 0, 5, \dots, 40\}$$

- waypoints ahead of car ($s_i \geq 0$)
- waypoint behind car ($s_i = -5$)
- track centerline
- ▲ car (pose = pos + yaw)